Generic PCM Days

Development of a new hydrological model for early Mars

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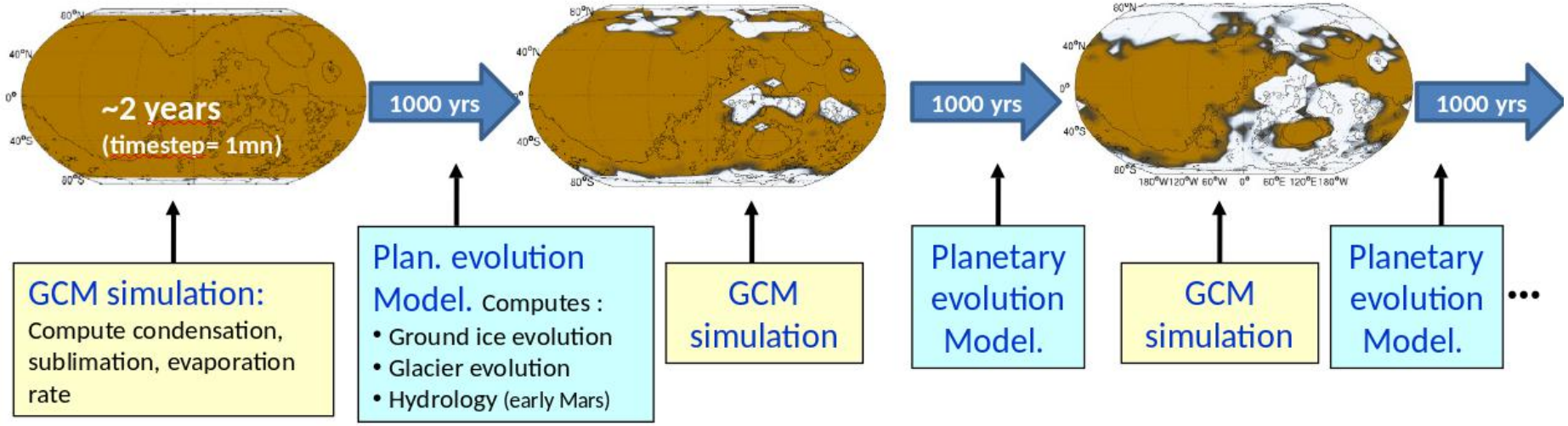
Planetary Evolution Model



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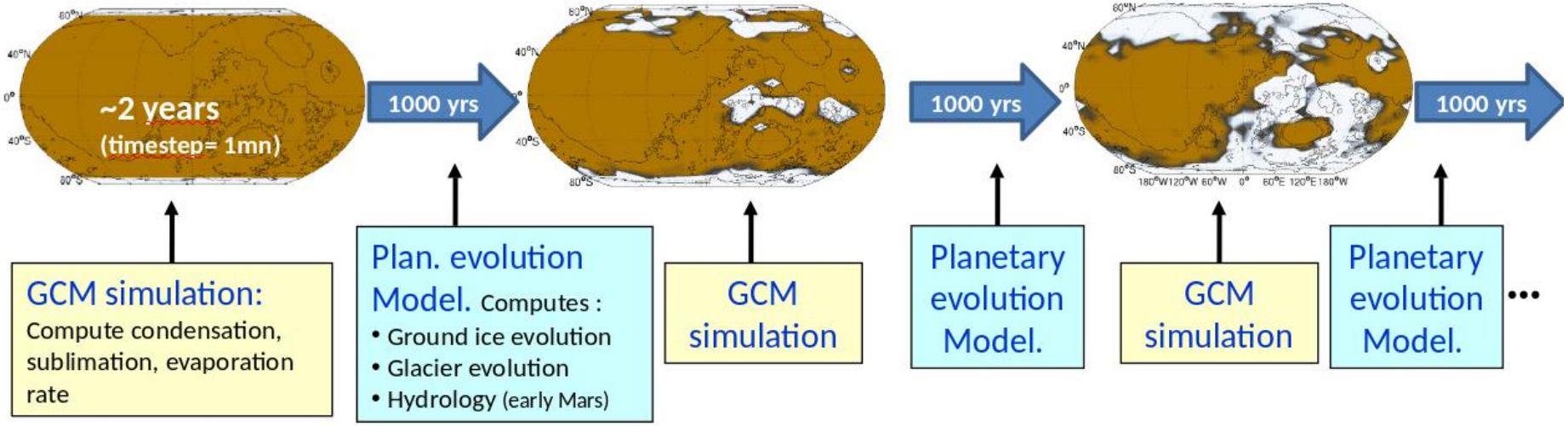
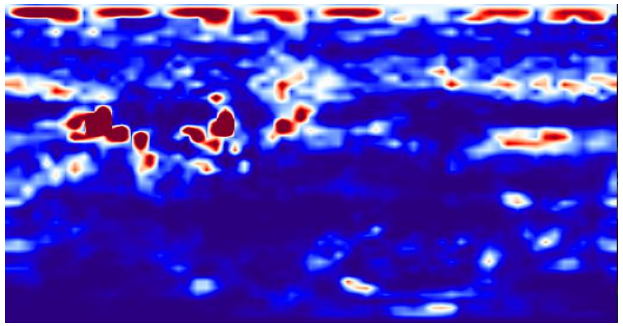


Precipitation/Evaporation rate map



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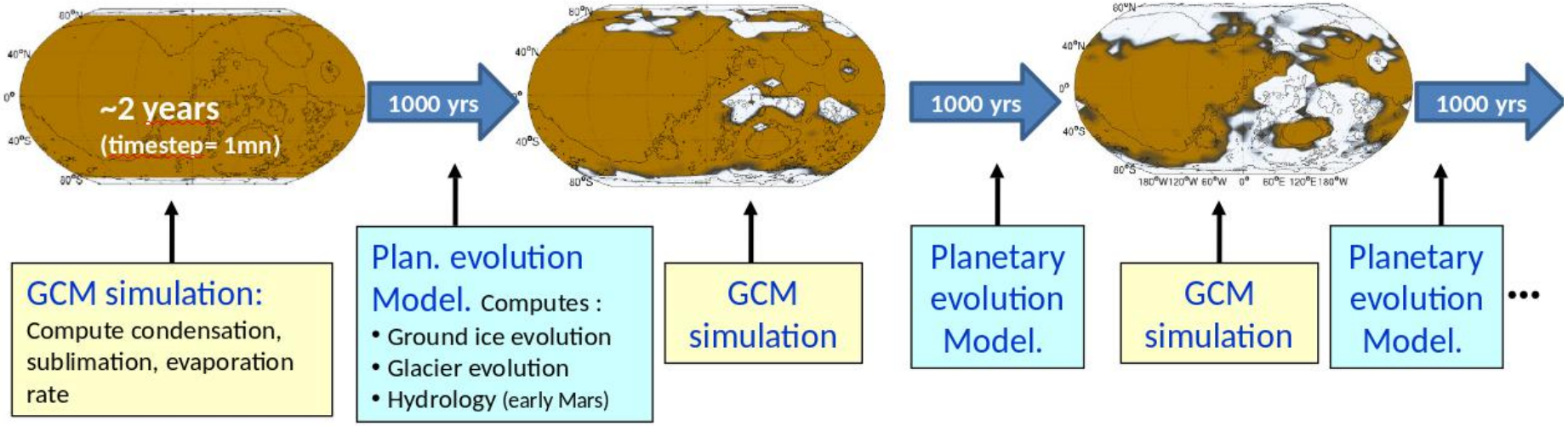

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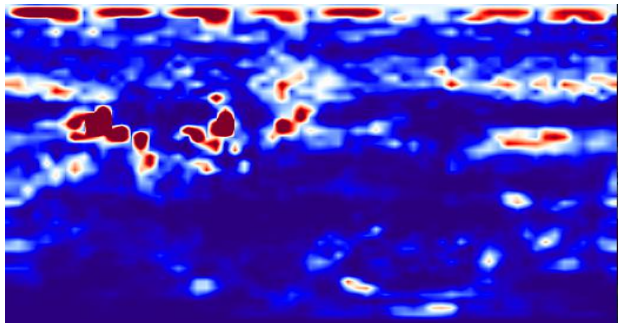
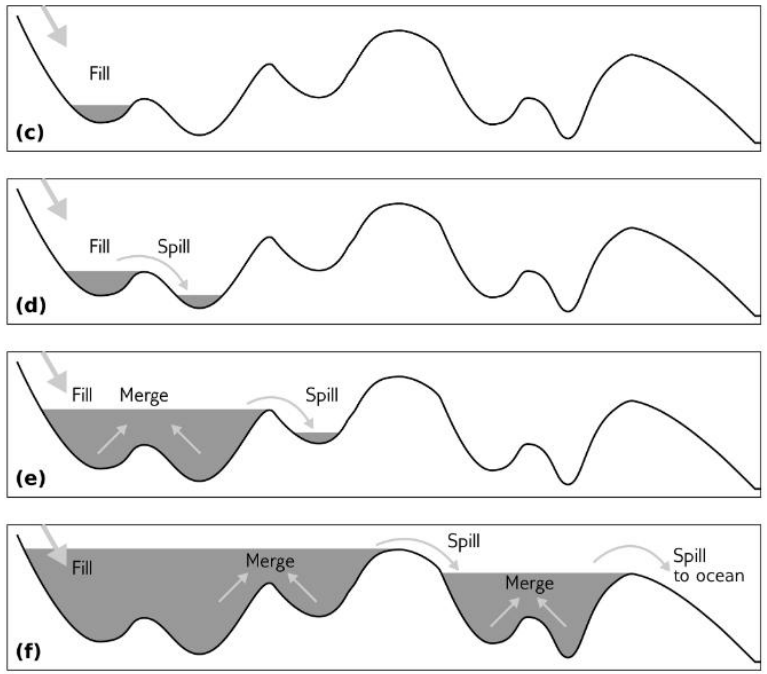


Hydrologic Methodology

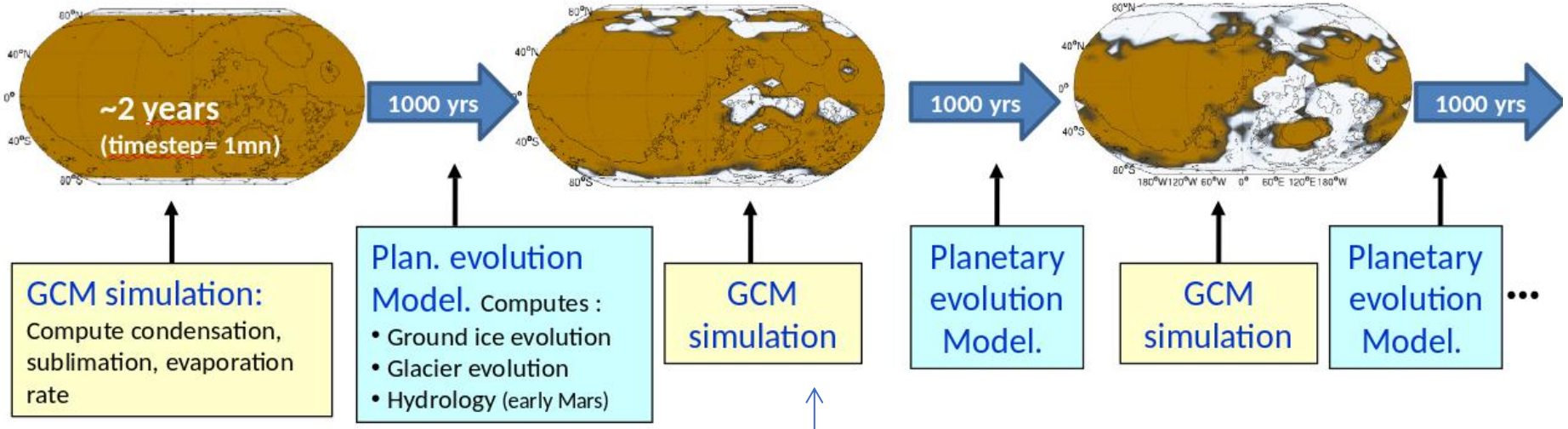


(Barnes et al., 2021)

2 steps:

- Hydrologic Database
- Hydrologic Model

Planetary Evolution Model



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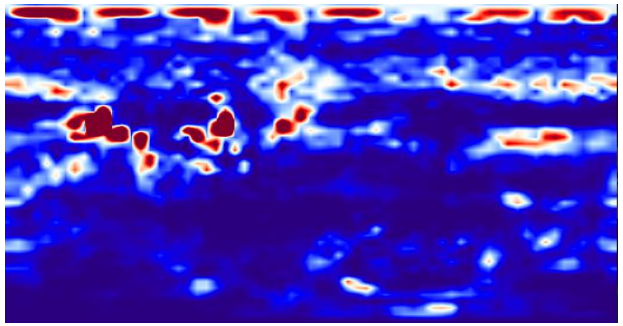

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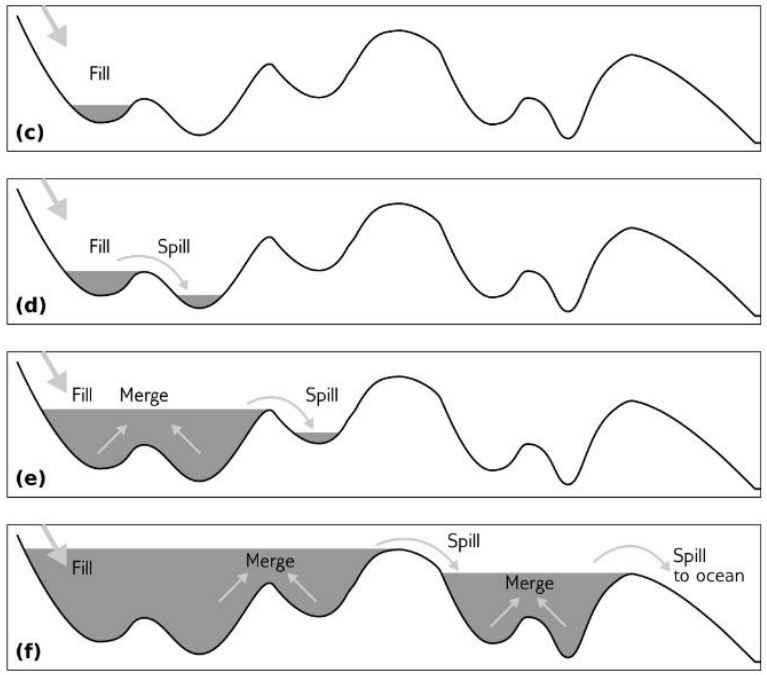
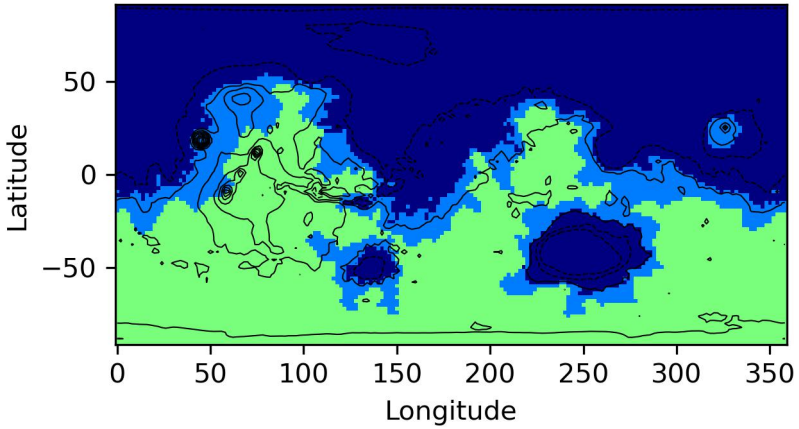


(Barnes et al., 2021)

Water surface map



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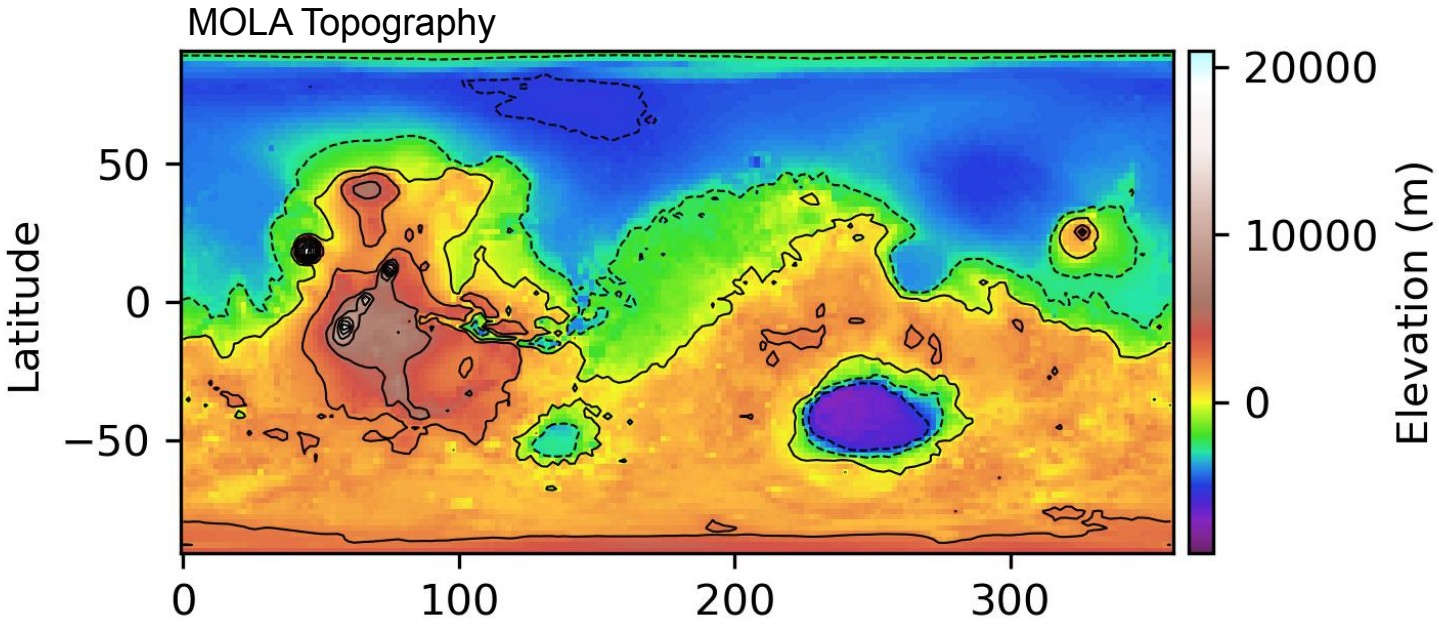
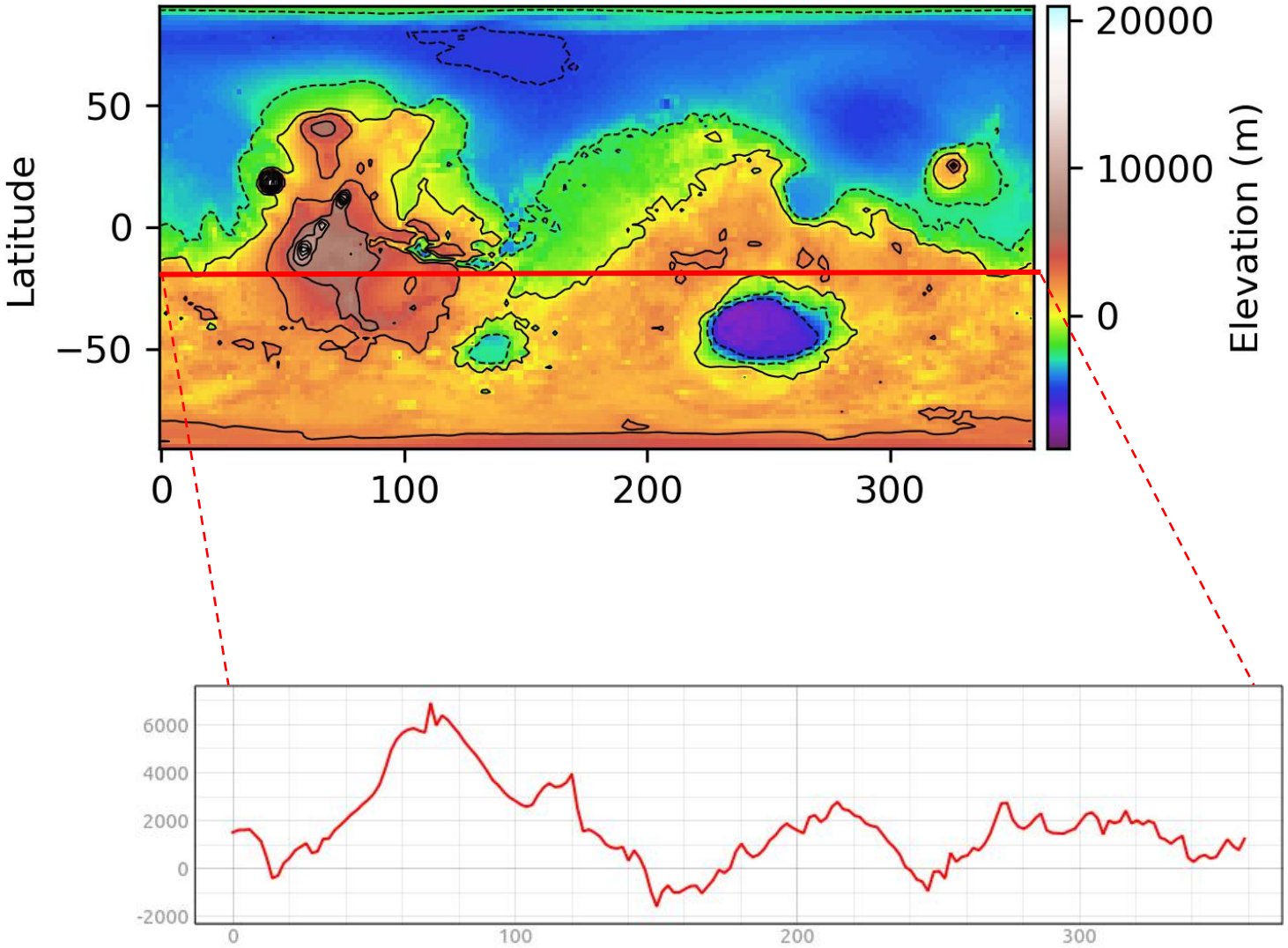
Hydrologic Database



Hydrologic Database

MOLA Topography



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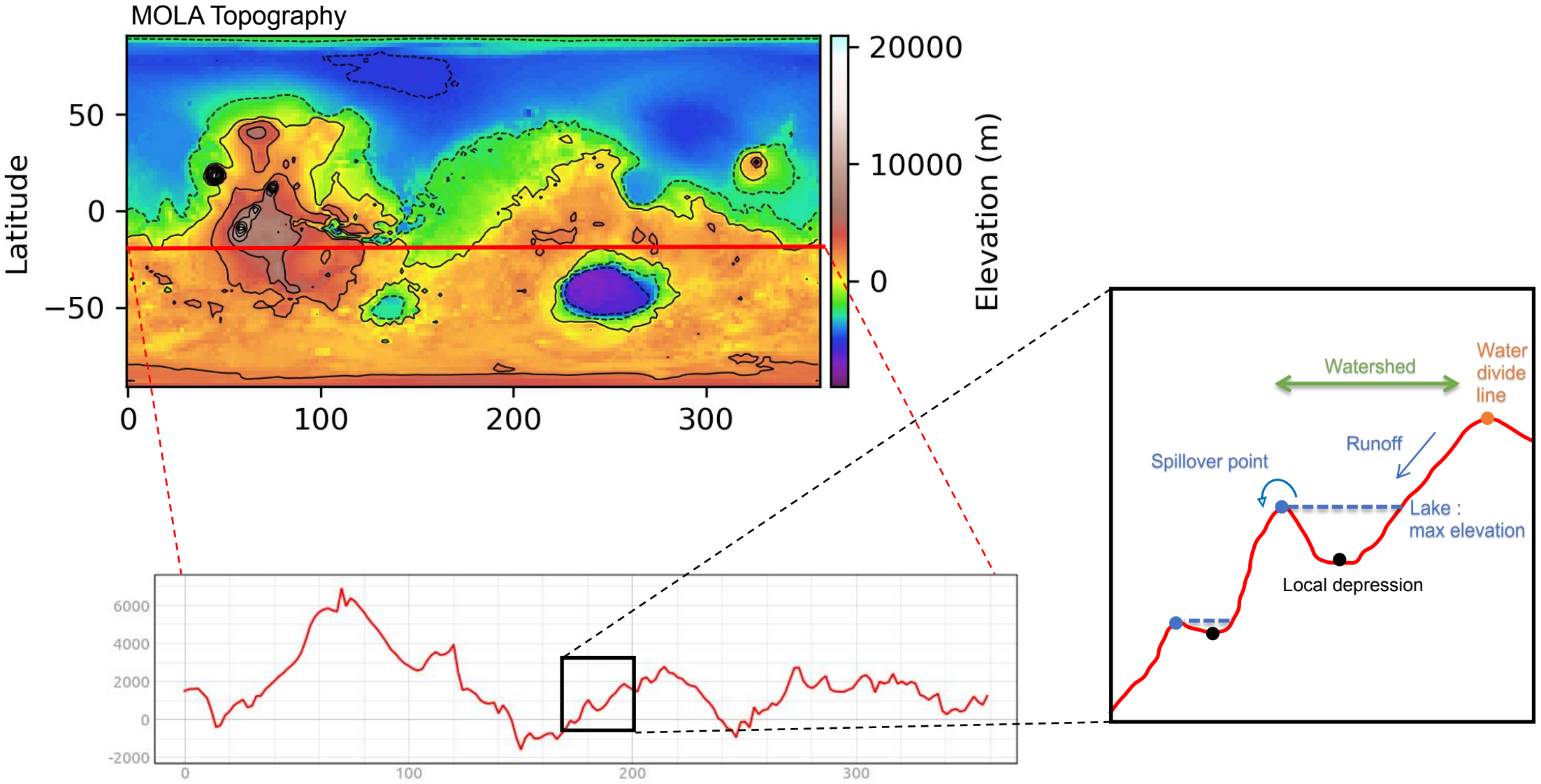

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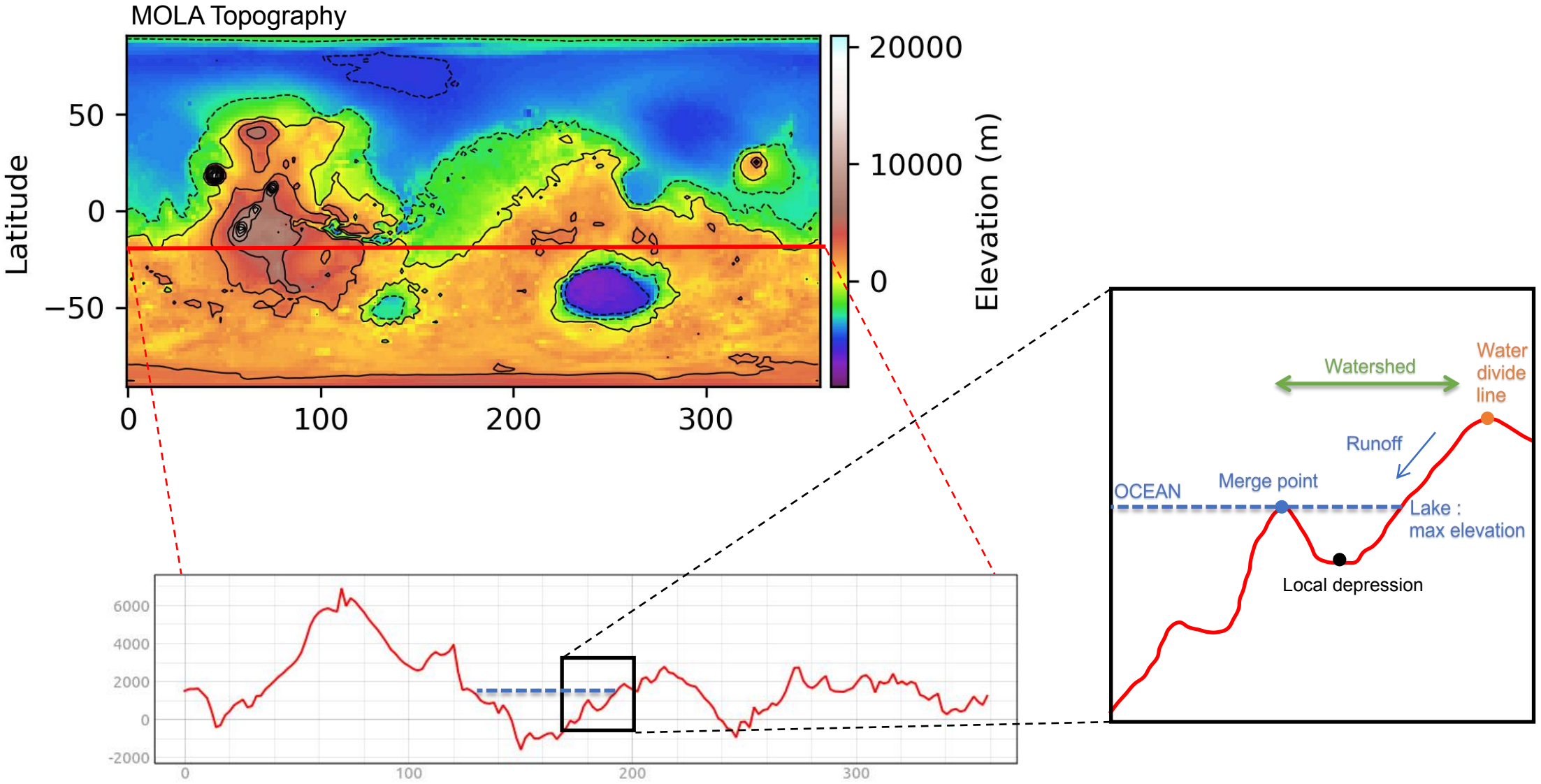

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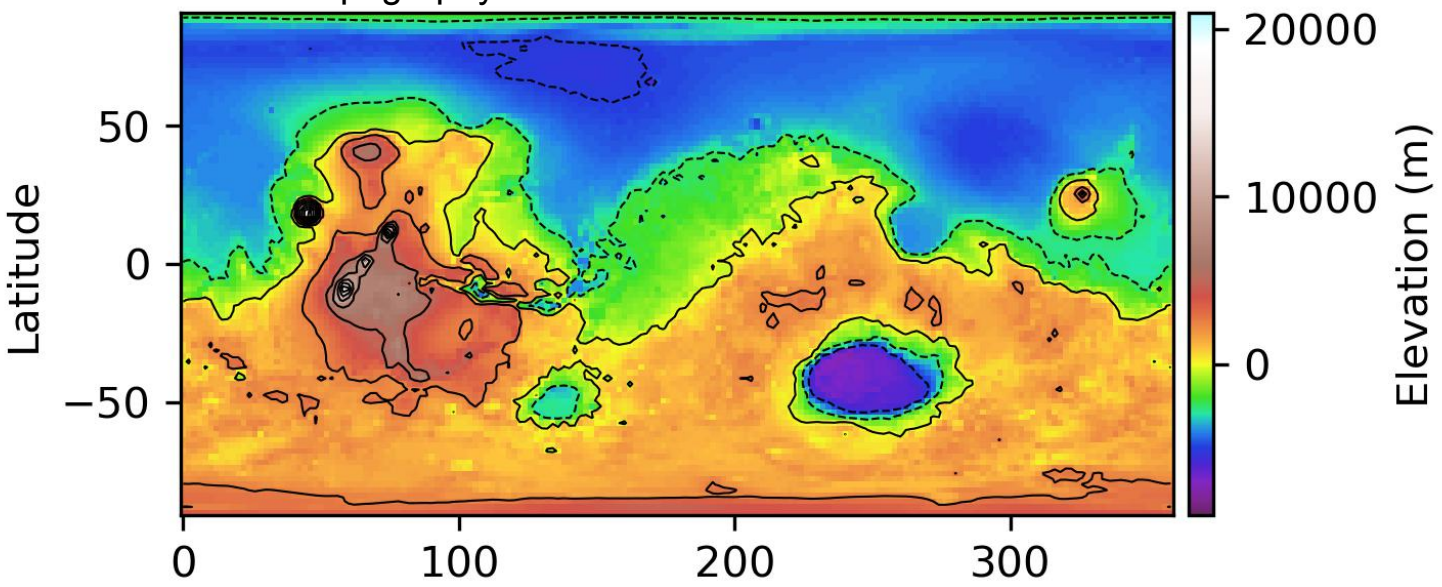

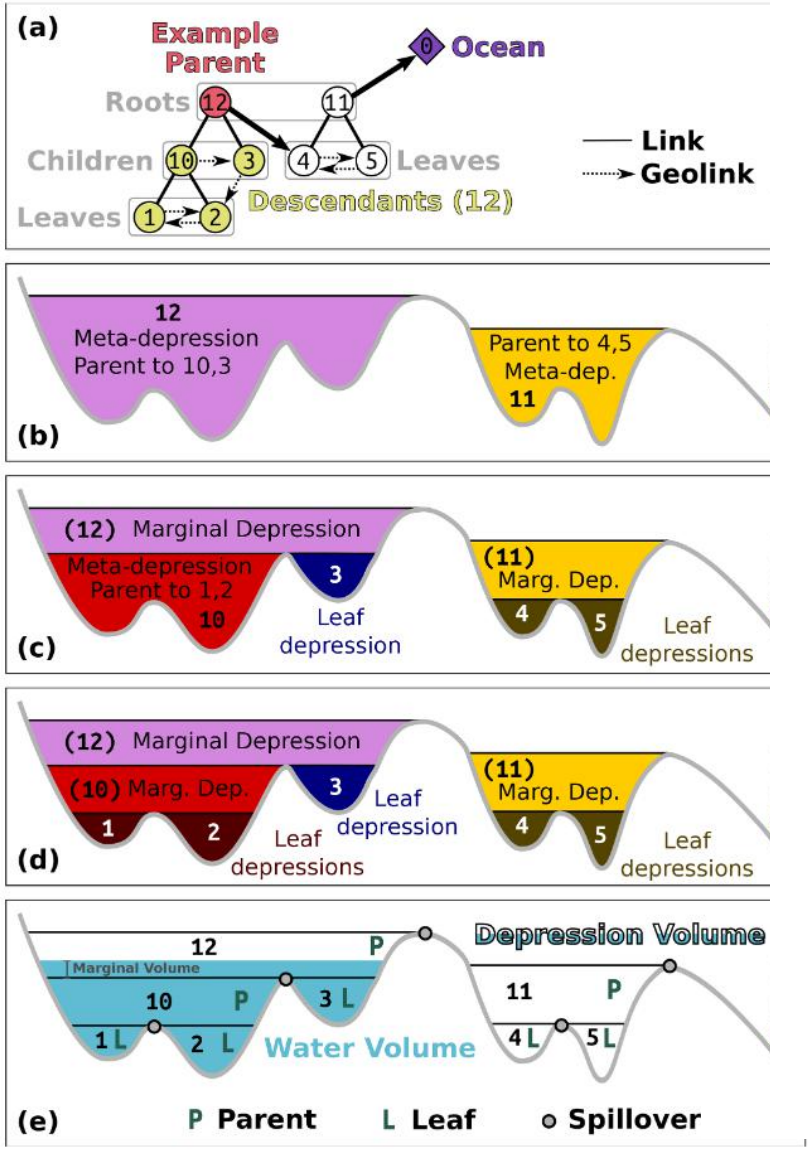

(Barnes et al., 2021)

Hydrologic Database

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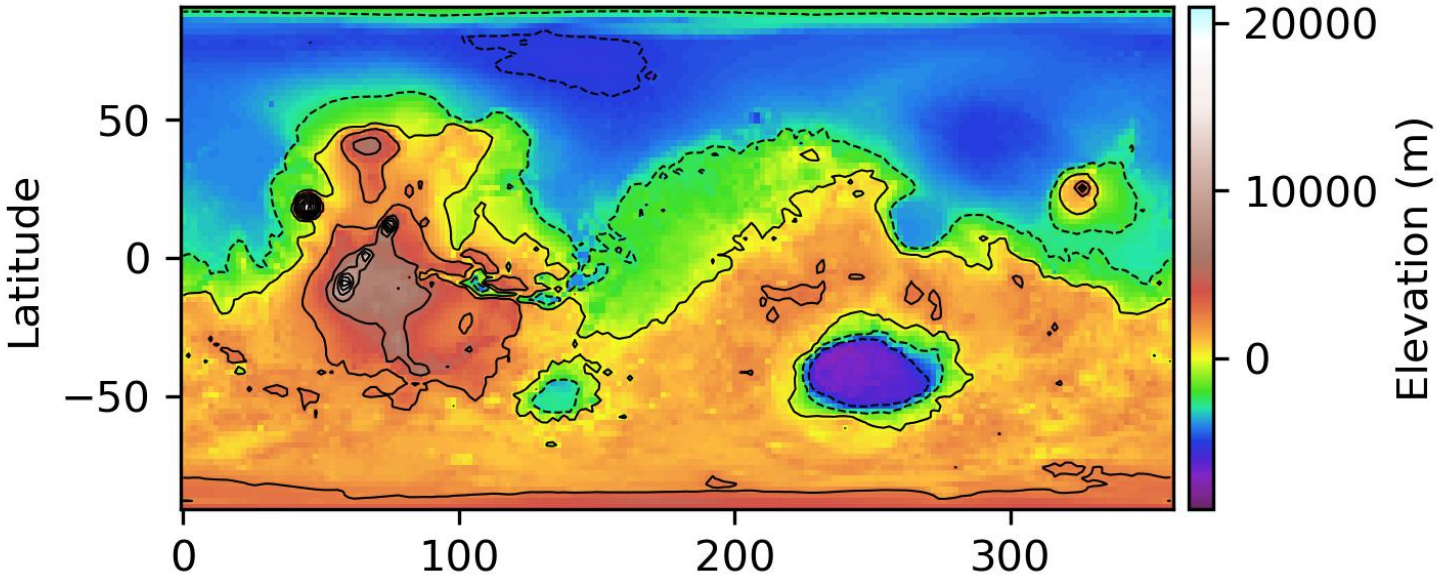
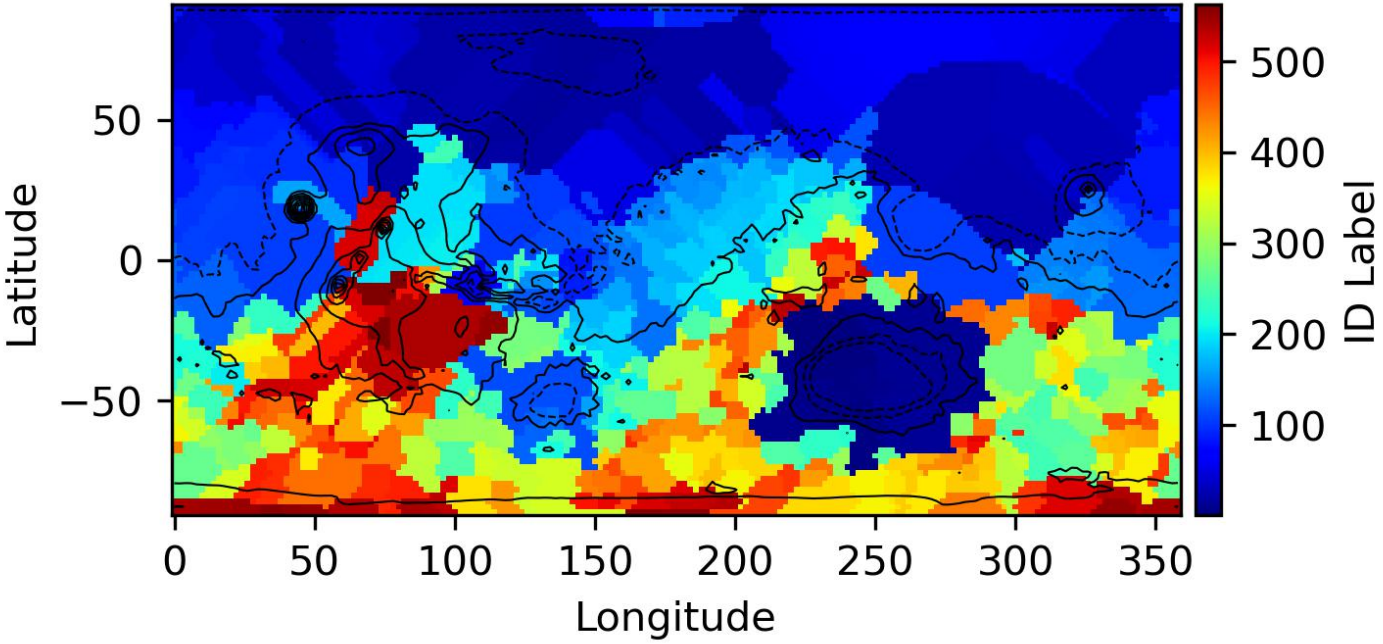
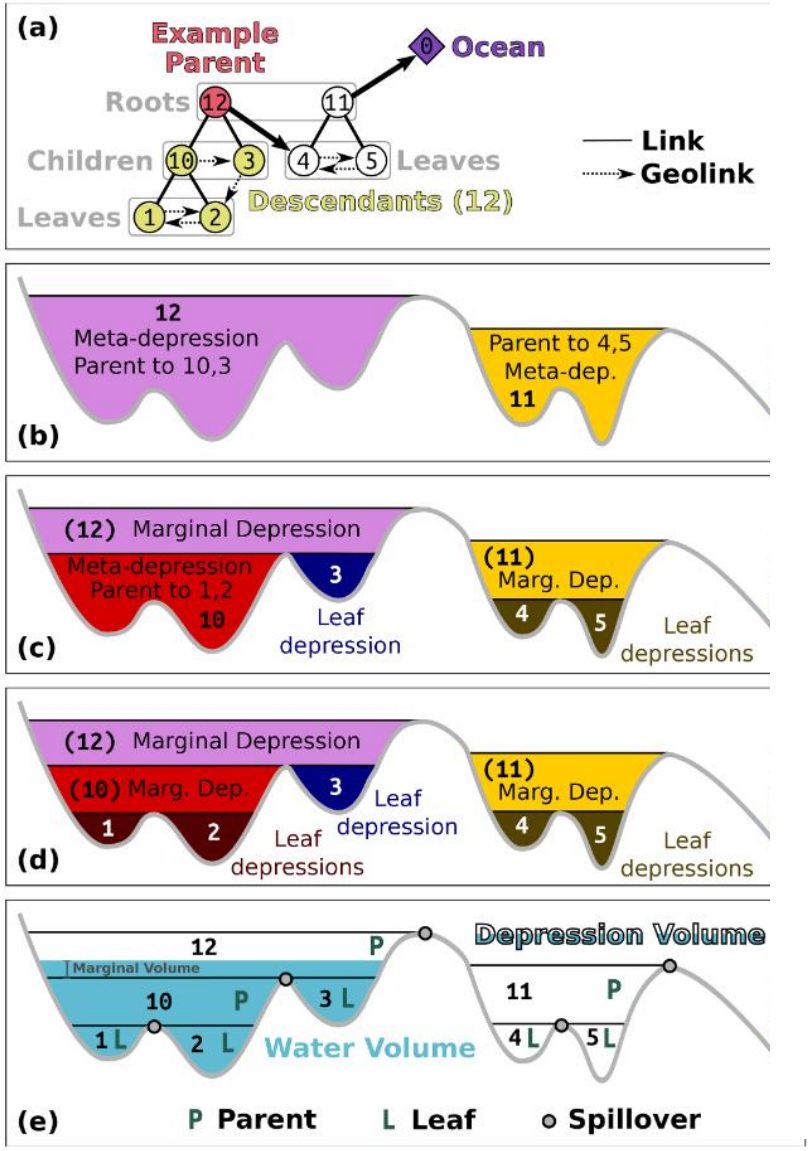


Watershed of leaf depressions





(Barnes et al., 2021)

Parameter [a]	Description	Unit
i	Depression ID (a child depression)	-
p_i	Parent depression ID of the depression i	-
s_i	Sibling depression ID of the depression i	-
g_i	First order depression ID where the depression i spill over	-
c_i^{pit}	Cell ID of the pit	-
c_i^{out}	Cell ID of the spillover point	-
Z_i^{max}	Elevation of the spillover point of the depression i	m
Z_i^{min}	Elevation of the spillover point of the two childs of the depression i	m
A_i	Watershed area of the depression i	m^2
V_i^{max}	Maximum water volume of the depression i	m^3
V_i^{min}	Minimum water volume of the depression i	m^3
λ_i^{lake}	Longitude of the bottom of the depression lake i	degrees (°)
φ_i^{lake}	Latitude of the bottom of the lake	degrees (°)
λ_i^{min}	Minimum longitude of the depression i	degrees (°)
λ_i^{max}	Maximum longitude of the depression i	degrees (°)
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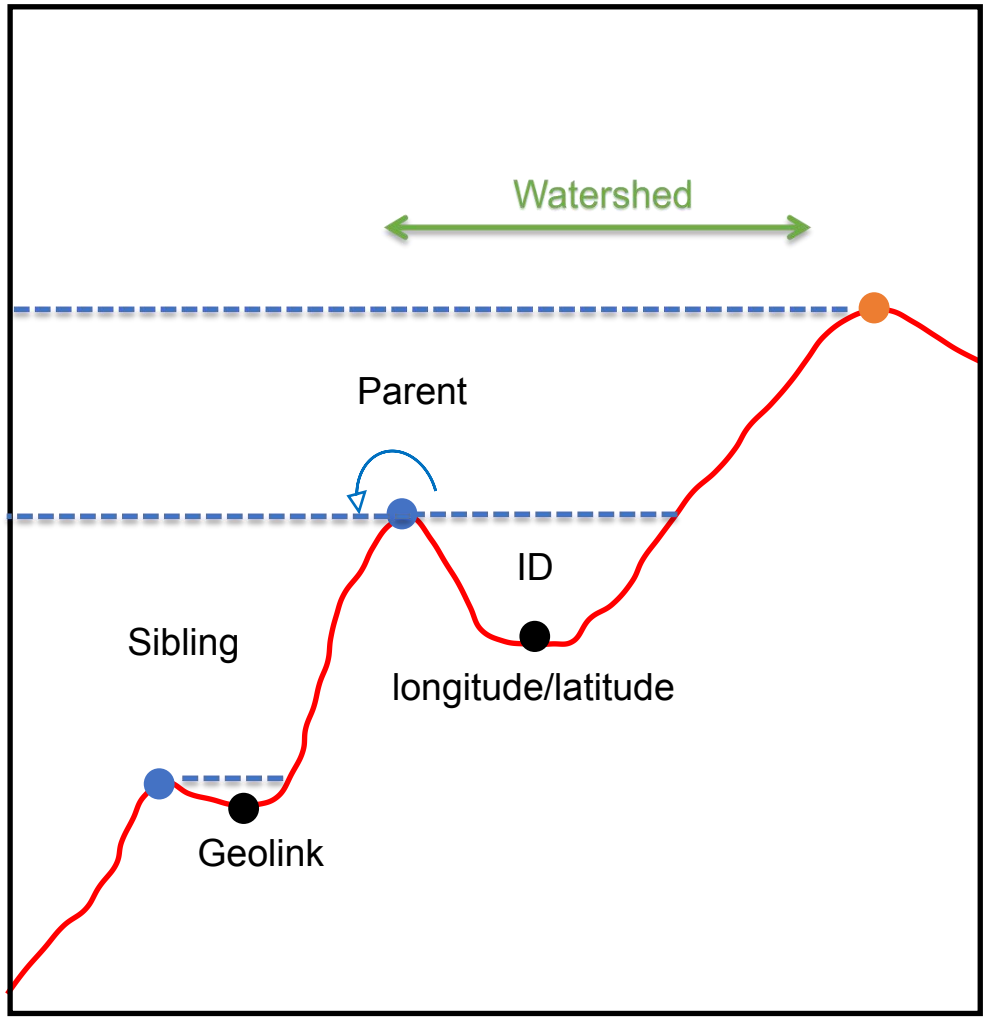

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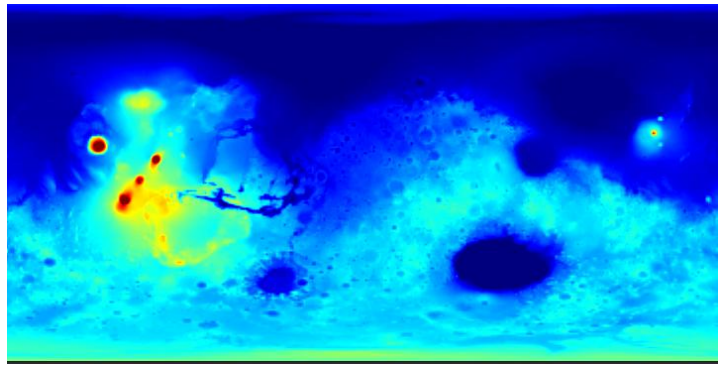
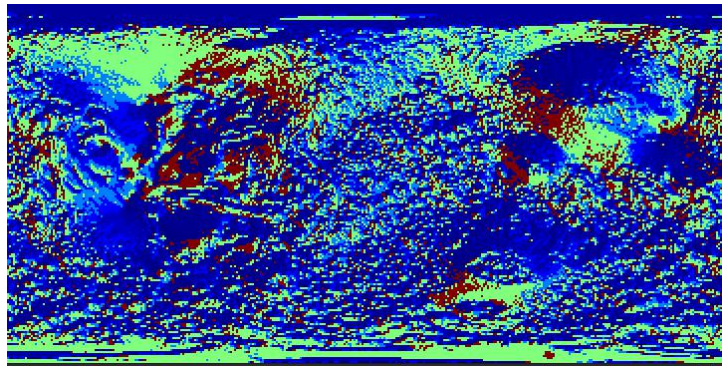
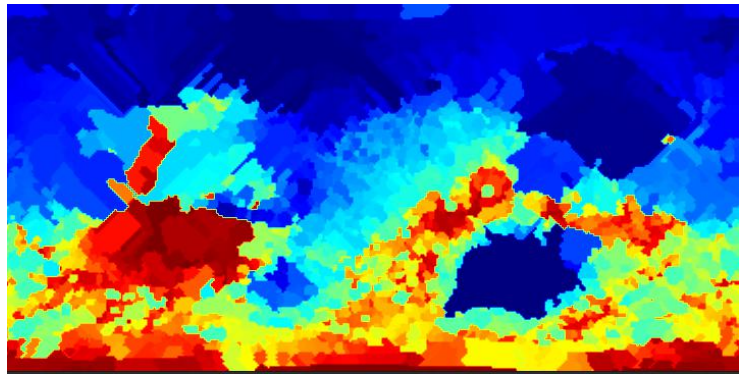
Elevation map



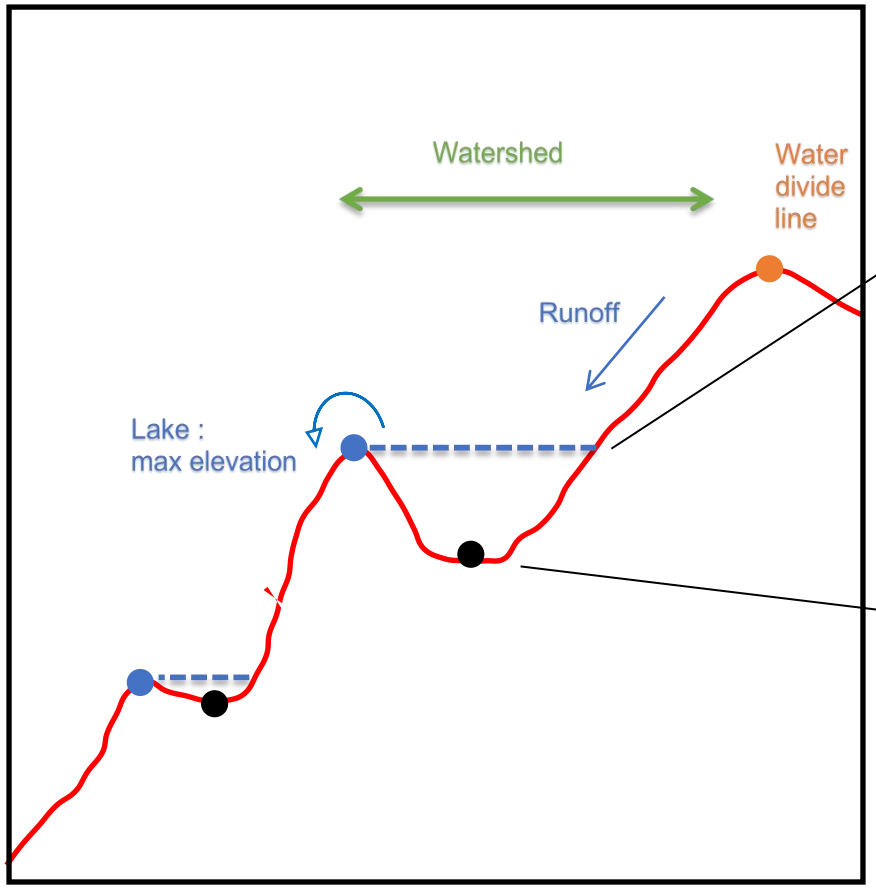
Flow direction map



Watershed Depression ID

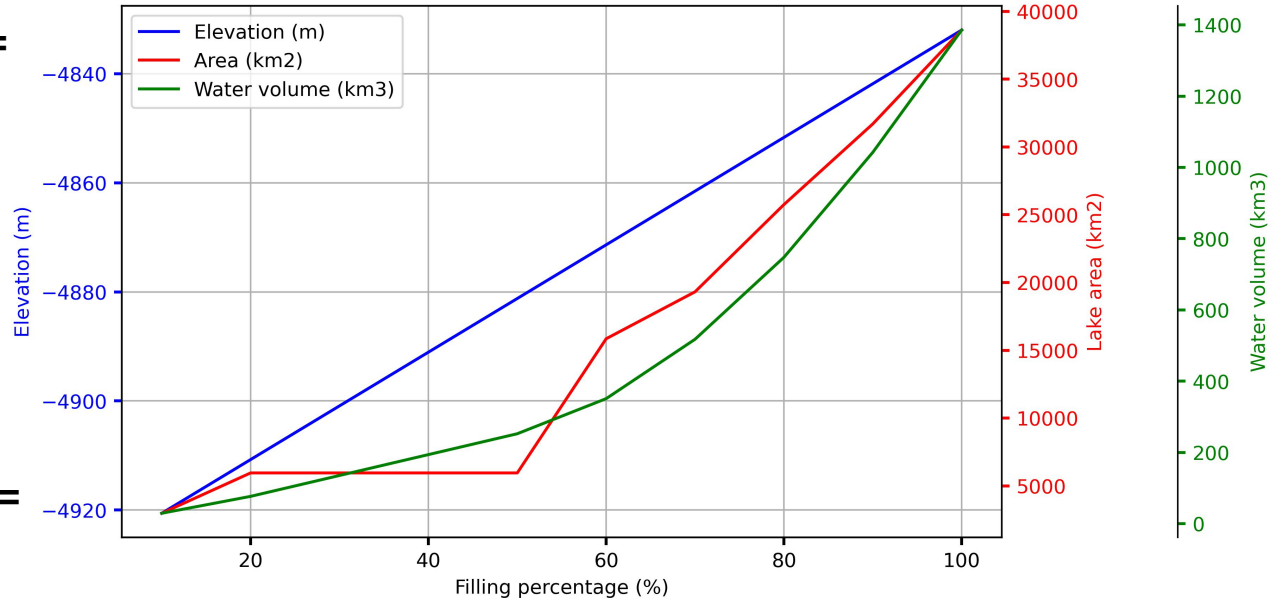


Hydrologic Database: relationship between lake volume, lake elevation and lake area

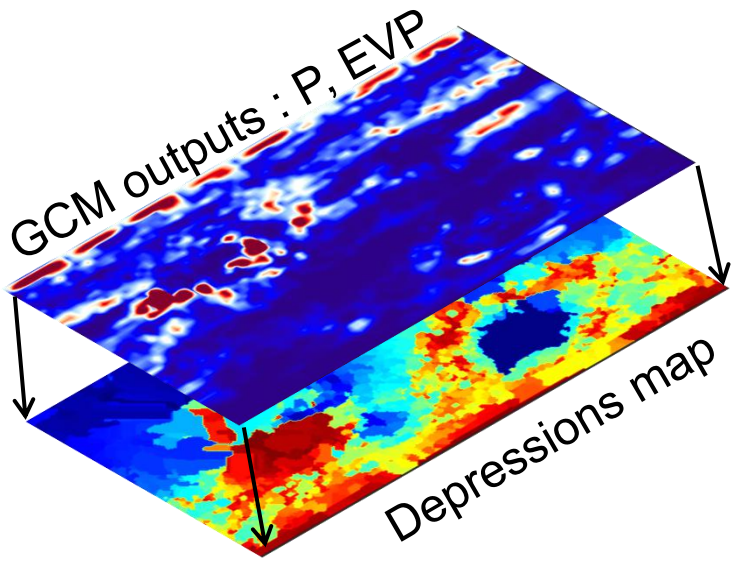


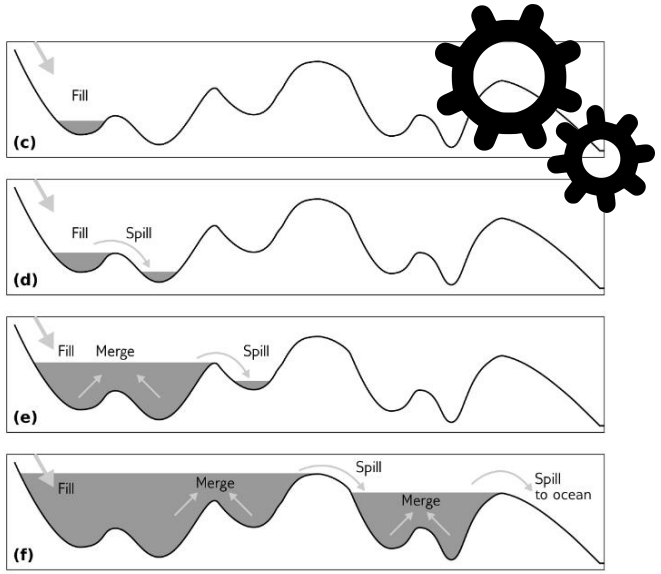
Z_{max} =

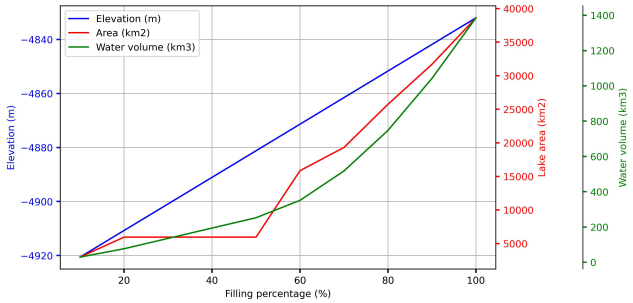
Z_{min} =



Hydrologic Model





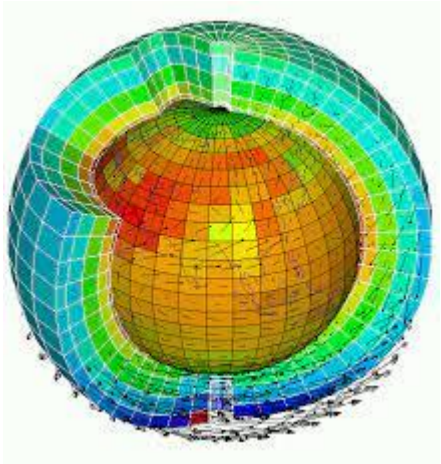


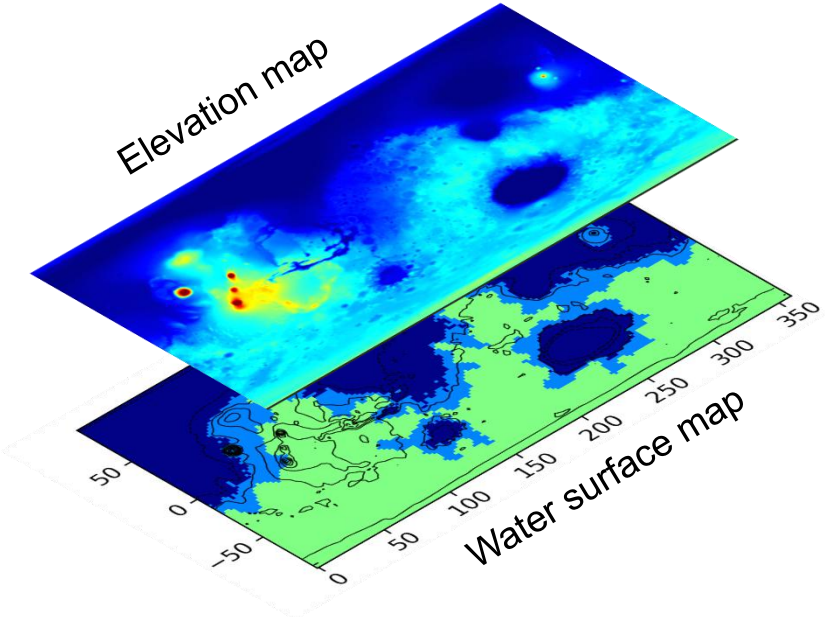
Hydrologic Database

Lake volume

↓

Lake elevation
Lake area





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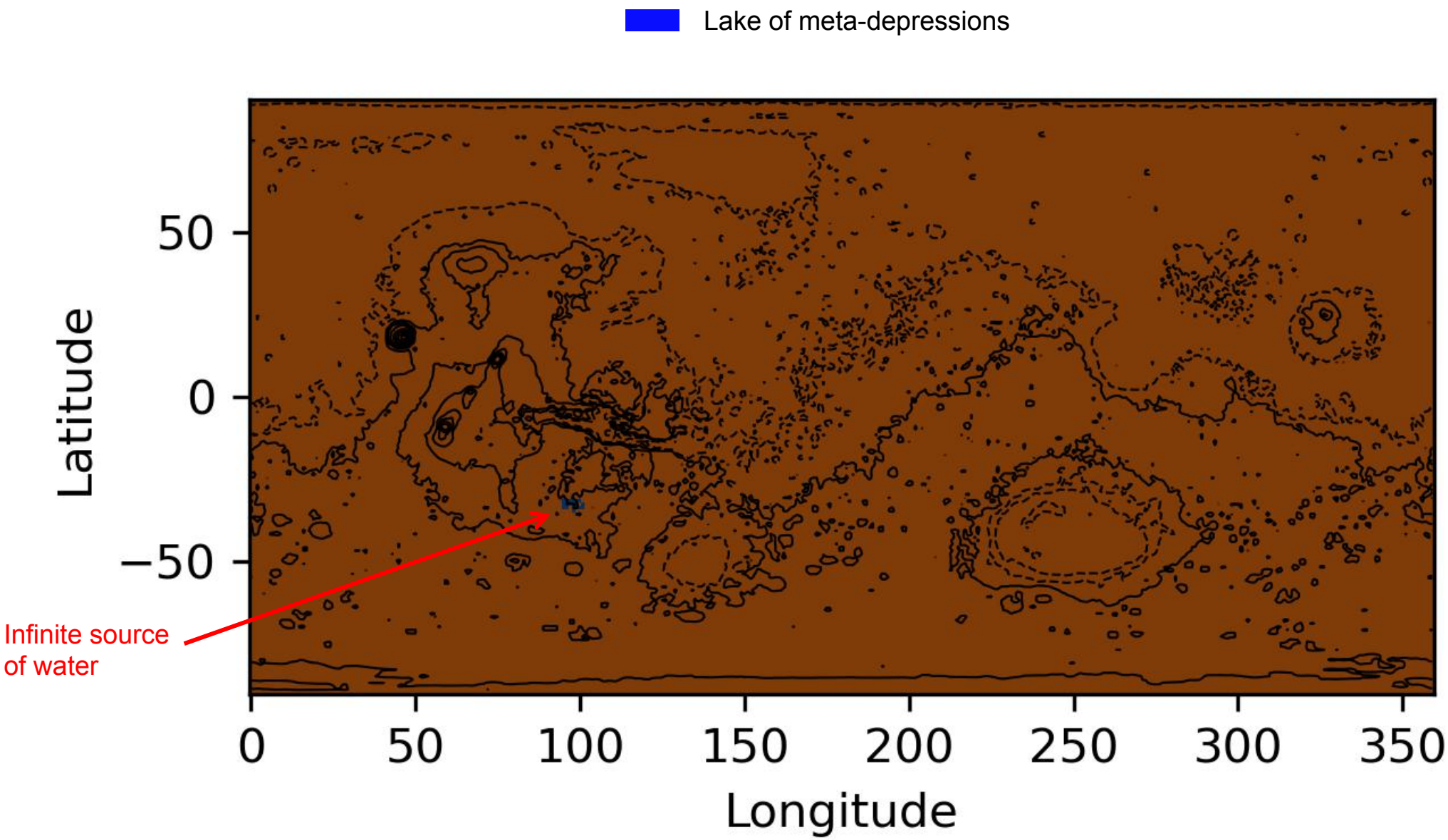
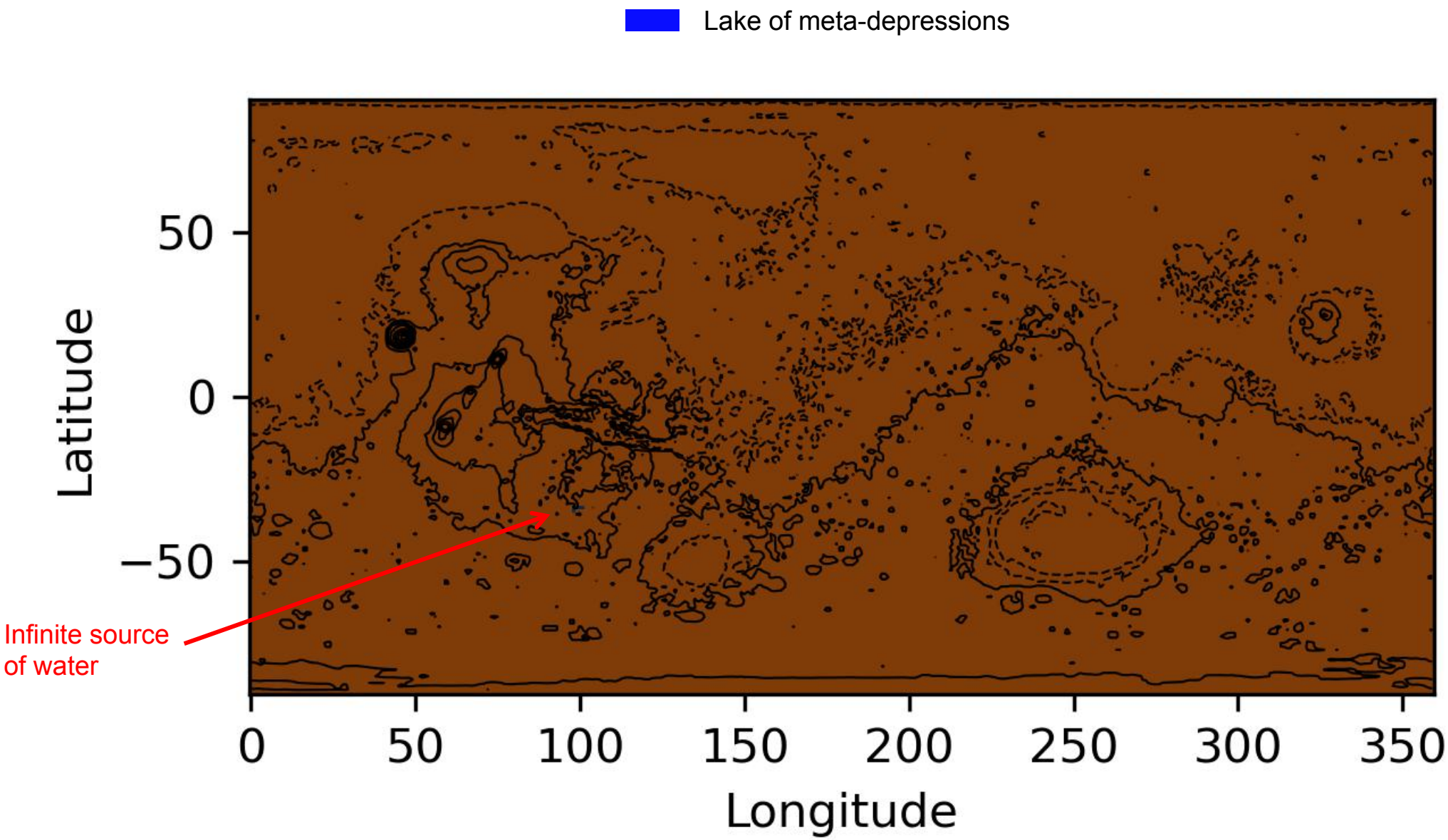
Hydrologic Database